INVESTIGATION OF NONLINEAR PROPERTIES OF ANIMAL SPINAL CORD IN THZ RANGE

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Abstract

In the present work the electrodynamic properties of a system consisting of animal spinal cord nerve fibres in the terahertz range (0.1 - 3 THz) are experimentally investigated using time domain spectrometer. It is shown that when an external constant voltage is applied to the sample, significant changes in the parameters of the spinal cord sample (refractive index, absorption) occur when the electric field vector of the terahertz radiation is oriented parallel to the nerve fibres.

A model was proposed, according to which under the influence of a voltage greater than threshold, the opening of the channels of nodes of Ranvier of the nerve fibres occurs, due to which the nerve fibres are transformed into specific wires with nonlinear conductivity, and the entire sample into a diffraction grating consisting of these wires. Computer modelling of the nonlinear interaction of THz radiation with the spinal cord sample was performed, the results of which are in good agreement with the experimentally obtained results.

Introduction

According to many studies (for example [1-3]), the exchange of information in living organisms is carried out by an electrochemical method, by the propagation of electrical pulses through nerve fibres. The structure of nerve fibres was revealed, models of the nervous system were built, and theories were created that successfully explain a number of processes, occurring in the nervous system. The speed of propagation of electrical pulses passing through nerve fibres was measured [4], and does not exceed ~120 m/s.

However, many processes occurring in the nervous system, such as information perception and analysis, memory, logical reasoning, learning, consciousness, emotions, etc., imply significantly higher rates of information exchange, and therefore cannot be explained within the framework of existing theories [5]. Indeed, during the last decades, various studies have been carried out on the recording of high-frequency processes occurring in the nervous system [6-8]. Signals in the GHz range (1.5 to 4.5 GHz) [9] and in the radio range (up to 30 MHz) [10] have been experimentally recorded.

Based on the structural features of nerve fibres and their size, it is assumed in a number of works that nerve fibres can serve as waveguides for exchanging information through THz or infrared electromagnetic waves. Various models have been proposed and various mechanisms of information exchange have been considered [11-15], but many processes occurring in the nervous system have not been clarified to date. Therefore, multivariate studies of neural networks can be very useful in revealing the mechanism of fast processes occurring in nervous systems.

Recently, some peculiarities of animal spinal cord in the THz range of electromagnetic waves have been revealed through our experimental research [16, 17]. In particular, it was shown that after some time of applying a voltage greater than a certain threshold value, the time domain shape of the THz pulse passed through the sample of the nerve fibre network, and therefore also the Fourier spectrum, changes significantly.

After applying the voltage, resonant absorption appears in the spectrum of the pulse passing through the sample, the resonance frequency of which begins to shift to the low-frequency range over time. In parallel, the high-frequency components of the spectrum grow significantly over time. After some time (around 5 minutes) this process stops, and after eliminating the voltage the THz pulse passing through the sample returns to its original form within about 1 hour.

In this work, the nonlinear properties of the nervous network of the animal's spinal cord in the terahertz range of electromagnetic waves were experimentally investigated under the influence of external voltage.

Experimental results and discussions

The measurement methodology is similar to that presented in [16, 17]. Flat-parallel samples (with dimensions 20x20x0.1 mm^3) from the white matter of the animal spinal cord (pig and sheep) were placed in cuvettes made of polytetrafluoroethylene (PTFE) plates which are transparent in terahertz range.

To study the behaviour of spinal cord nerve tissues in the THz range under conditions of applied external voltage, tungsten electrodes were mounted in a cuvette at a distance of around 15 mm from each other parallel to nerve fibres. Studies were conducted on experimental setup assembled on the basis of Menlo Systems TERA K15 THz time-domain spectrometer.

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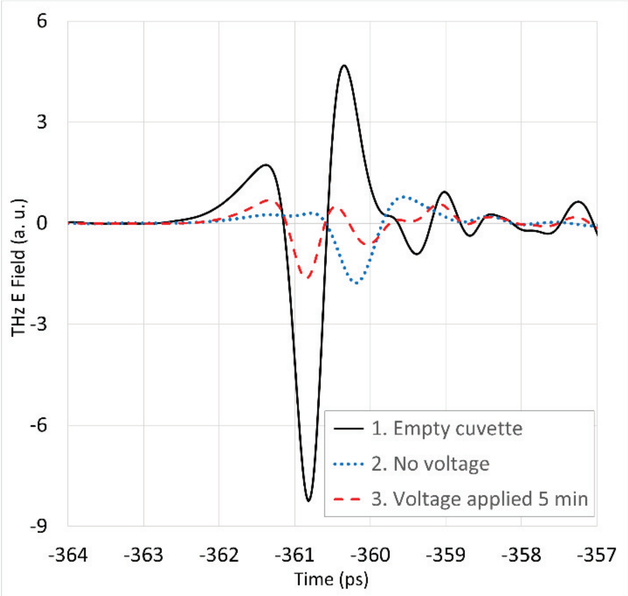
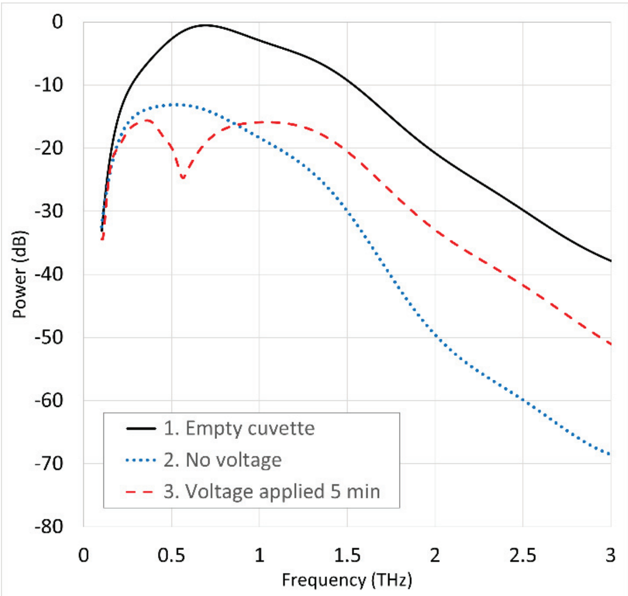
A specially developed computer program was used to process and analyse the measurement results.



a



b

Fig. 1 (a) The time-domain profiles of THz pulses: waveform 1 - black (solid). the pulse transmitted through empty cuvette, waveform 2 - blue (dotted). through the cuvette containing spinal cord sample without external voltage, waveform 3 - red (dashed). through the same sample 5 minutes after applying 100 V voltage. (b) The corresponding power spectra of the pulses shown in Fig. 1(a)

Fig. 1a shows the time-domain profiles of THz pulses: waveform 1 - black (solid) shows the pulse passing through an empty cuvette, waveform 2 - blue (dotted) through a cuvette containing a 0.1 mm thick spinal cord sample without external voltage, and waveform 3 - red (dashed) through the same sample 5 minutes after a 100 V voltage was applied, with the electric field of THz pulse aligned along the nerve fibres. Fig. 1b shows the corresponding power spectra of the THz pulses from Fig. 1a.

It can be seen from Fig. 1, that in the THz range, the spinal cord sample has quite strong absorption in the absence of voltage. The absorption of a layer with a thickness of 0.1 mm is around 5 dB at a frequency of 0.25 THz, it increases with increasing frequency, and approaches 30 dB at 2 THz. In the low-frequency region of the THz range, the refractive index is close to 4 in the absence of voltage and decreases with increasing frequency, approaching approximately 3.6 (as shown by waveform 1 in Fig. 2). Notably, spinal cord samples from different animals showed no significant differences and exhibited similar parameters, regardless of their orientation to the plane of polarization of the THz radiation. It can also be seen from Fig. 1 that the parameters of the spinal cord sample undergo significant changes over time when applying voltage. The shape of the pulse is greatly changed while there is no significant change in the average power. The power spectrum also undergoes a sharp change (see waveform 3, Fig. 1(b)). After 5 minutes of applying voltage to the sample, a strong resonance absorption is observed near 0.6 THz, and the high-frequency components of the spectrum increase significantly starting from approximately 0.9 THz.

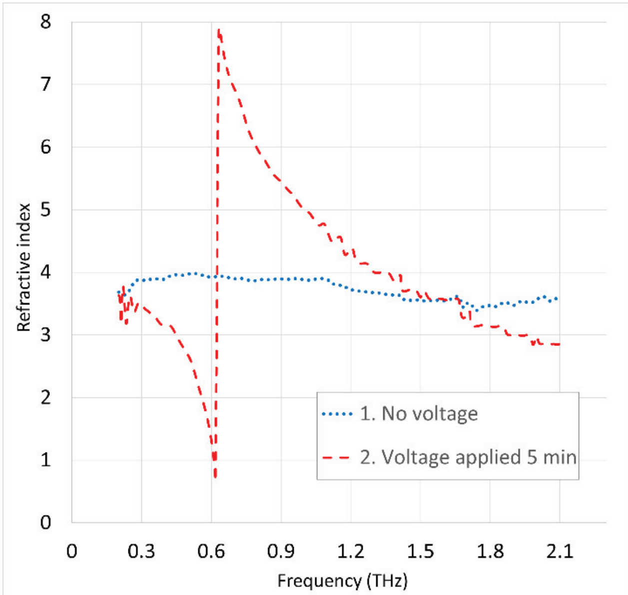


Fig. 2 Waveform 1 - blue (dotted) - Dependence of refractive index on frequency in the absence of voltage, waveform 2 - red (dashed) after 5 minutes of applying voltage to the sample.

Additionally, the THz pulse begins to outpace the pulse obtained without external voltage, see Fig. 1(a), which is equivalent to the decrease in the refractive index of the sample. However, the distortion of the pulse shape suggests a strong dependence of the refractive index on frequency. Based on a detailed analysis of the phase spectrum of the THz pulse passing through the sample, the dependence of refractive index on frequency was obtained (waveform 2 in Fig. 2).

Note that for this spinal cord sample, near the resonant frequency, the refractive index value becomes less than one. This fact, as well as the fact that the amplitudes of the high-frequency components of the THz pulse spectrum increase when passing through the sample taken from the spinal cord after applying voltage, allows us to conclude that in the presence of voltage, the examined sample turns into a network of wires parallel to the electric field of the THz pulse, which are also endowed with non-linear conductivity.

In this case, resonances and generation of combination frequencies of the spectral components of the THz pulse may occur in the system. Particularly, some spectral components may be generated inside the sample and reradiated earlier than ones with the same frequencies which are merely transmitted through the sample. Therefore, there may be an impression that some spectral components of the THz pulse propagate faster through the filled cuvette than through the empty cuvette, meaning, they seem to propagate at a speed greater than the speed of light.

To further ensure that the animal's spinal cord behaves as a non-linear system, low-pass filter was put together with the spinal cord sample, so that the THz pulse propagates through both of them. Firstly, the filter was placed before the sample, and in the second case - after the sample. In the first case, the high-frequency components of the recorded pulse were larger than in the second case, which proves that high-frequency components are really generated in the sample.

In addition, a computer simulation for the generation of combination frequencies was also performed, accepting the investigated sample as a system of wires, in which the dependence of the current on the electric field of the pulse is non-linear. That dependence was approximated by a power polynomial:

$$J = a_1 E + a_2 E^2 + a_3 E^3 + a_4 E^4 + a_5 E^5$$

The THz pulse spectrum obtained as a result of modelling, in the case of the following polynomial coefficients values a1=0.075, a2=0.56, a3=0.224, a4=-0.14, a5=-0.09, is shown in Fig. 3 (waveform 3). For comparison, the experimentally obtained results in the absence of voltage (waveform 1) and 2 minutes after applying the voltage of 100 V (waveform 2) are also shown in the same figure. It can be seen from the figure that waveforms 2 and 3 are similar in nature.

Note that despite the fact that the parameters of different samples taken from the spinal cord differ from each other when voltage is applied, all samples demonstrate similar patterns. After the voltage is removed, the parameters of the samples are restored after about 1 hour.

It should also be noted that when the nerve fibres are positioned in a direction perpendicular to the electric field of the THz pulse, even when applying sufficiently high voltages, no noticeable changes in the absorption and refractive index of the sample occur.

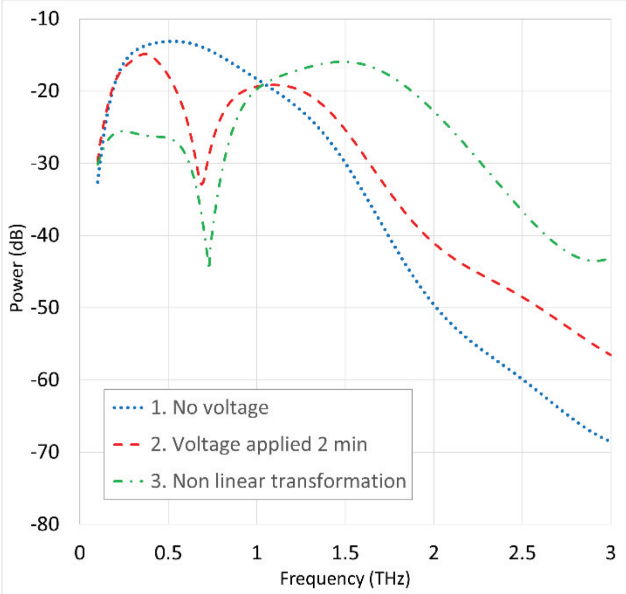


Fig. 3 Comparison of power spectra of THz pulse passing through the sample and its nonlinear transformation: waveform 1 - blue (dotted) - power spectrum of THz pulse passed through the sample without external voltage, waveform 2 - red (dashed) - with 100V applied to the sample for 5 min, waveform 3 - green (dash dotted) - result of modelling.

Thus, the analysis of the obtained results allows us to conclude that under the influence of a sufficiently high voltage, apparently, the opening of the channels in the nodes of Ranvier of the nerve fibres occurs, due to which the nerve fibres are transformed into specific wires with nonlinear conductivity, and the entire sample into a diffraction grating consisting of these wires.

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